

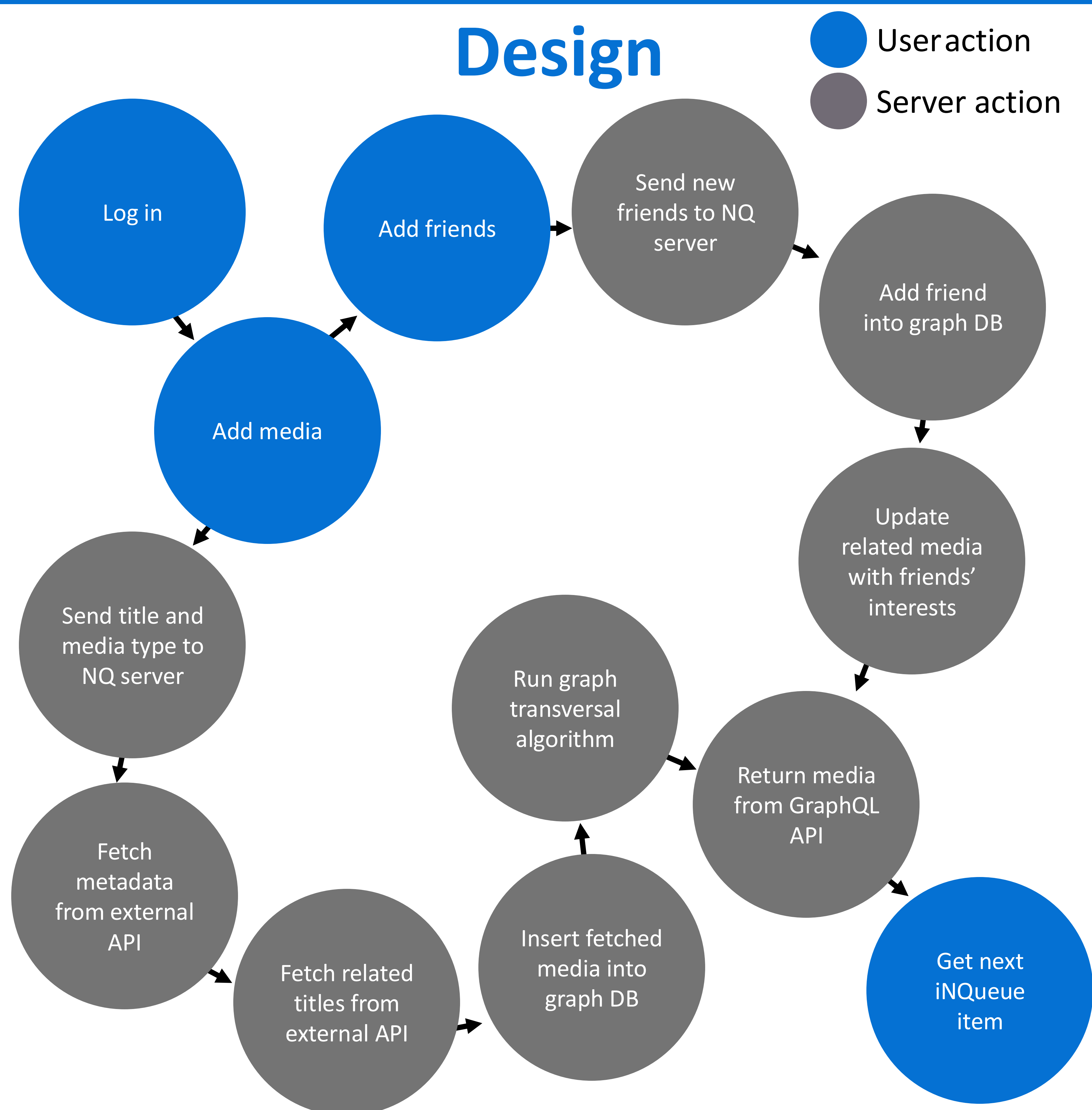
Abstract

NQ is a cross-platform media recommendation system that unifies user preferences across all media formats to deliver personalized "next in queue" suggestions. By aggregating and restructuring data from multiple APIs, NQ creates a graph of your media consumption history.

Features

- User Experience: Native mobile capabilities create more intuitive interactions that improve usability for diverse populations
- Scalable: Architecture can easily support more nodes
- Open Ecosystem: Built on open-source technologies, enabling community contributions and transparent development

Design



Merits

- Brings together complex data from several sources
- Hard to create relationships without a good ingestion pipeline
- Normalizing different media types (games, movies, books) presents a tough challenge
- Making the input and recommendation requests responsive requires precise data requests

Challenges

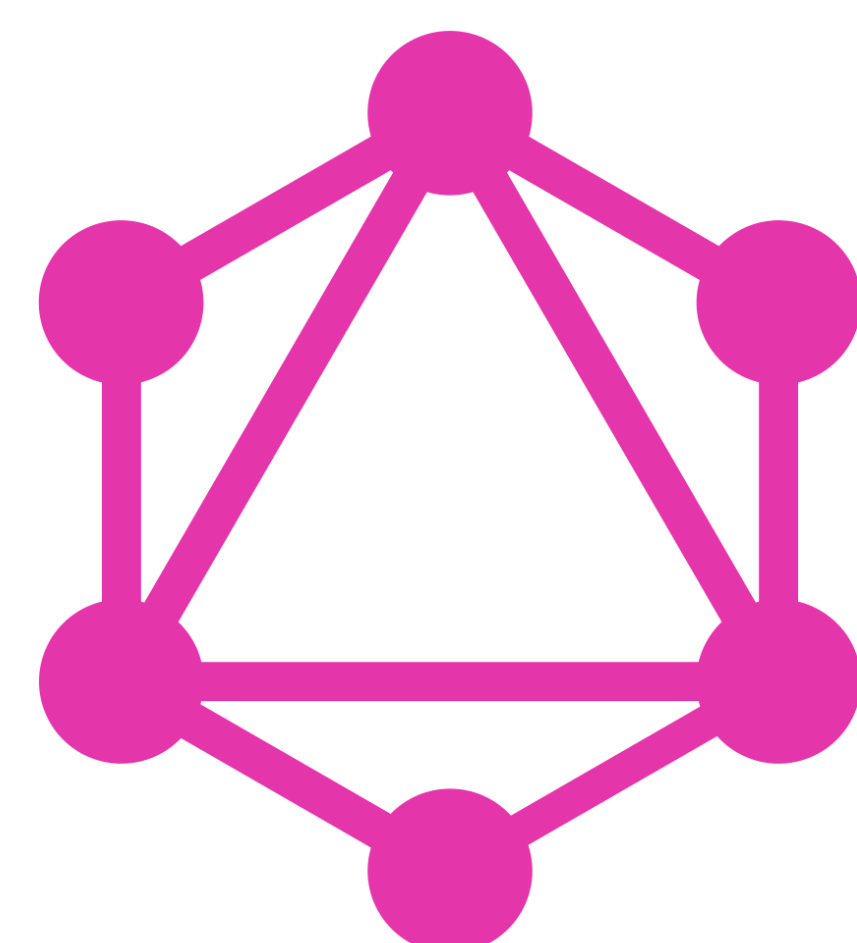
- Limited access to automatically integrating history (Spotify, YouTube, Netflix)
- Different APIs provide wildly different quantities and quality of data
- New to GraphQL, Neo4j, and React Native

Technologies

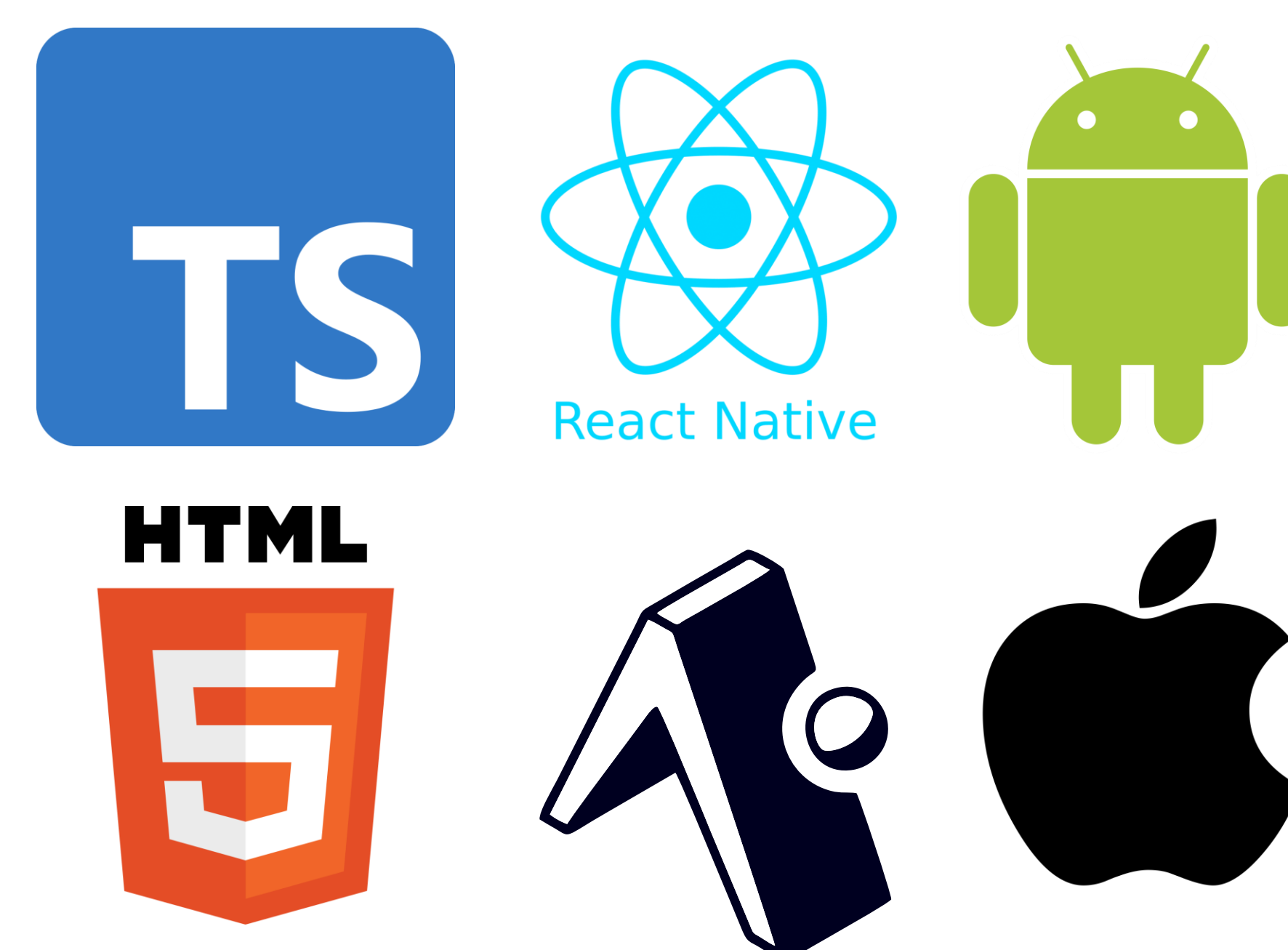
Backend



API



Frontend



Future

- Deploy to App Store and Google Play Store
- Add support for connecting music and games
- Allow users to connect to sources for automatic tracking (Spotify, Goodreads, Letterboxd)
- Improve app performance and load times
- Fine tune recommendation algorithm to improve precision and accuracy based on user input